Composite Products

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
Wood Plastic Composite Decking; Wood Plastic Composite Cladding	IS 17420 (Wood Plastic Composite Products for Building Applications); IS 4984; IS 12818; IS 14885; IS 14925; IS 4985; IS 7834; IS 9537; IS 12231; IS 13592; IS 14182; IS 14333; IS 15265; IS 15328; IS 15778; IS 16088	ISO 16616 (Test Methods for Wood-Plastic Composites — WPC Deck Boards); ISO 20819 (Wood-Plastic Composites — Specifications)	ASTM D7032 (Establishing Performance Ratings for Wood-Plastic Composite Deck Boards and Guardrail Systems); ASTM D7031 (Evaluating Mechanical and Physical Properties of Wood-Plastic Composite Products); ASTM D790 (Flexural Properties of Plastics); ASTM D638 (Tensile Properties of Plastics); ASTM G154 (UV Exposure of Nonmetallic Materials)	DIN EN 15534-1 (Wood-Plastic Composites — Characteristics of WPC Products); DIN EN 15534-4 (Decking Profiles and Tiles); DIN EN ISO 527-1 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN ISO 4892-3 (Artificial Weathering)	ICC-ES Acceptance Criteria (Building Products); ASTM-based systems excluded; UL 94 (Fire Classification, where applicable); GREENGUARD Certification (Indoor Applications)
WPC Door Frame	IS 17420 (Wood Plastic Composite Products for Building Applications); IS 4984; IS 12818; IS 14885; IS 14925; IS 4985; IS 7834; IS 9537; IS 12231; IS 13592; IS 14182; IS 14333; IS 15265; IS 15328; IS 15778; IS 16088	ISO 16616 (Wood-Plastic Composites — Test Methods); ISO 20819 (Wood-Plastic Composites — Specifications)	ASTM D7031 (Mechanical and Physical Properties of Wood-Plastic Composite Products); ASTM D638 (Tensile Properties of Plastics); ASTM D790 (Flexural Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics)	DIN EN 15534-1 (Wood-Plastic Composites — General Requirements); DIN EN ISO 527-1 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN ISO 4892-3 (Artificial Weathering); DIN EN 1534 (Determination of Resistance to Indentation, where applicable)	ICC-ES Evaluation Criteria (Building Components); GREENGUARD Certification (Indoor Air Quality Applications)
FRP Pultruded Profile	IS 6746 (FRP Structural Profiles for Building/Industrial Applications);IS 4984; IS 12818; IS 14885; IS 14925; IS 4985; IS 7834; IS 9537; IS 12231; IS 13592; IS 14182; IS 14333; IS 15265; IS 15328; IS 15778; IS 16088	ISO 14125 (Fibre-reinforced Plastic Composites — Determination of Flexural Properties); ISO 527-5 (Plastics — Determination of Tensile Properties — Unidirectional Fibre-reinforced Plastic Composites)	ASTM D3917 (Dimensional Tolerances of Thermosetting Glass-Reinforced Plastic Pultruded Shapes); ASTM D638 (Tensile Properties of Plastics); ASTM D790 (Flexural Properties of Plastics); ASTM D2584 (Ignition Loss of Cured Reinforced Resins); ASTM D4099S (Standard Specification for Poly(Vinyl Chloride) (PVC) Prime Windows/Sliding Glass Doors and Profile Extrusions); ASTM D5260S (Standard Classification for Chemical Resistance of Poly(Vinyl Chloride) (PVC) Homopolymer and Copolymer Compounds and Profile Extrusions); ASTM D1042S (Standard Test Method for Linear Dimensional Changes of Plastics Caused by Exposure to Heat and Moisture (Profile Stability)); ASTM D256S (Standard Test Methods for Determining the Izod Pendulum Impact Resistance of Plastics (Applicable to profile section testing))	DIN EN 13706-1 (Reinforced Plastics Composites — Pultruded Profiles); DIN EN 13706-2 (General Specifications); DIN EN ISO 527-4 (Tensile Properties of Fibre-Reinforced Plastics); DIN EN ISO 14125 (Flexural Properties of Fibre-Reinforced Composite Materials)	ANSI/ACMA FRP Standards; OSHA Structural Safety Requirements; NEMA FG Standards (Electrical Applications)
GFRP Rebar	IS 18256 (Glass Fibre Reinforced Polymer Reinforcing Bars)	ISO 10406-1 (Fibre-reinforced Polymer (FRP) Reinforcement of Concrete — Test Methods — FRP Bars and Grids); ISO 18319 (Fibre-	ASTM D7957 (Solid Round Glass Fiber Reinforced Polymer Bars for Concrete Reinforcement); ASTM D7205 (Tensile Properties of Fiber Reinforced Polymer	DIN EN ISO 10406-1 (Fibre-Reinforced Polymer (FRP) Reinforcement for Concrete — Test Methods); DIN EN ISO 527-4	ACI 440 Guidelines (FRP Reinforced Concrete Applications); AASHTO LRFD Bridge Design Guidelines (FRP Reinforcement

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
		reinforced Polymer Reinforcement for Concrete — Requirements)	Matrix Composite Bars); ASTM D7913 (Bond Strength of Fiber Reinforced Polymer Matrix Composite Bars); ASTM D7508 (Glass Fiber Reinforced Polymer Bars, where applicable)	(Tensile Properties of Fibre-Reinforced Plastics); DIN EN ISO 14125 (Flexural Properties); DIN EN ISO 4892-3 (Artificial Weathering, where applicable)	Applications)
GFRP Grating	IS 17431 (FRP Structural Profiles and Gratings)	ISO 14125 (Fibre-reinforced Plastic Composites — Determination of Flexural Properties); ISO 527-5 (Tensile Properties of Fibre-reinforced Composites)	ASTM D7290 (Determining Material Properties of Fiberglass Reinforced Plastic Composites); ASTM D635 (Flammability of Plastics); ASTM D790 (Flexural Properties of Plastics); ASTM E84 (Surface Burning Characteristics of Building Materials)	DIN EN ISO 14125 (Fibre-Reinforced Plastic Composites — Flexural Properties); DIN EN ISO 527-4 (Tensile Properties); DIN EN ISO 179-1 (Impact Strength); DIN EN ISO 4892-3 (Weathering Resistance)	OSHA Industrial Flooring Requirements; ANSI/NAAMM Standards for Metal/Composite Access Systems (where applicable); NSF/ANSI 61 (Water Applications)
FRP Cable Tray	IS 17431 (FRP Structural Profiles); IS 14772 (Cable Support Systems Applications); IS 17290; IS 16568; IS 17409; IS 17616	ISO 14125 (Fibre-reinforced Plastic Composites — Flexural Properties); ISO 527-5 (Tensile Properties of Fibre-reinforced Composites)	ASTM D635 (Flammability of Plastics); ASTM D638 (Tensile Properties of Plastics); ASTM D790 (Flexural Properties of Plastics); ASTM E84 (Surface Burning Characteristics of Building Materials)	DIN EN 61537 (Cable Tray Systems and Cable Ladder Systems for Cable Management); DIN EN ISO 14125 (Flexural Properties of FRP Composites); DIN EN ISO 527-4 (Tensile Properties); DIN EN ISO 4892-3 (Artificial Weathering)	NEMA FG 1 (Fiberglass Cable Tray Systems); UL 568 (Nonmetallic Cable Tray Systems); NFPA 70 (Electrical Installations)
FRP Ladder	IS 17431 (FRP Structural Profiles); IS 3696 (Ladders – Application)	ISO 14122-4 (Safety of Machinery — Permanent Means of Access — Fixed Ladders); ISO 14125 (FRP Composite Flexural Properties)	ASTM D635 (Flammability of Plastics); ASTM D638 (Tensile Properties of Plastics); ASTM D790 (Flexural Properties of Plastics); ASTM F1989 (Non-Destructive Testing of FRP Products, where applicable)	DIN EN 131 (Ladders — Requirements, Tests, Marking); DIN EN ISO 14125 (Flexural Properties of FRP Composites); DIN EN ISO 527-4 (Tensile Properties); DIN EN ISO 179-1 (Impact Strength)	ANSI A14 Series (Ladder Safety Requirements); OSHA Ladder Requirements; UL 94 (Electrical Insulating Applications, where applicable)
FRP Manhole Cover	IS 1726 (Manhole Covers and Frames)	ISO 15398 (Thermoplastics Manhole Covers and Frames for Underground Applications, where applicable); ISO 14125 (FRP Composite Properties)	ASTM D790 (Flexural Properties of Plastics); ASTM D638 (Tensile Properties of Plastics); ASTM D695 (Compressive Properties of Rigid Plastics); ASTM D2584 (Ignition Loss of Cured Reinforced Resins)	DIN EN 124-1 (Gully Tops and Manhole Tops — Definitions, Classification and General Principles); DIN EN ISO 14125 (Flexural Properties); DIN EN ISO 527-4 (Tensile Properties); DIN EN ISO 179-1 (Impact Strength)	AASHTO M306 (Drainage Structure Components); H20 Load Rating Requirements; EN/Utility Authority Specifications
Sandwich Composite Panel	IS 17384 (Sandwich Panels for Building Applications)	ISO 12623 (Thermal Insulating Products for Building Applications — Determination of Thickness of Pipe Insulation Products); ISO 7892 (Vertical Building Elements — Impact Resistance Tests)	ASTM C393 (Core Shear Properties of Sandwich Constructions by Beam Flexure); ASTM C365 (Flatwise Compressive Properties of Sandwich Cores); ASTM C297 (Flatwise Tensile Strength of Sandwich Constructions); ASTM D1781 (Climbing Drum Peel Test for Sandwich Panels)	DIN EN 14509 (Self-Supporting Double Skin Metal Faced Insulating Panels — Factory Made Products, where applicable); DIN EN ISO 14125 (Flexural Properties of FRP Composites); DIN EN ISO 527-4 (Tensile Properties); DIN EN ISO 1922 (Rigid Cellular Plastics — Shear Strength, where core applies)	NFPA 285 (Fire Propagation Test for Exterior Walls); FM Global Approval Standards (Building Panels); UL 790 (Roof Coverings, where applicable)

Source field: Plastic and Plastic Products